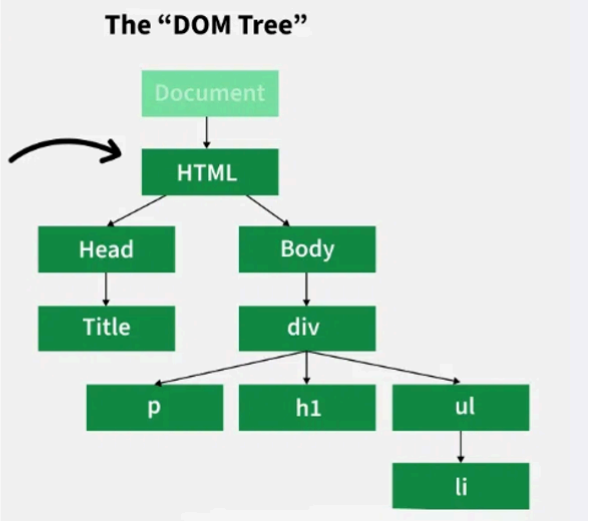
6. Understanding of DOM and Virtual DOM

▼ What is DOM?

- **DOM** stands for **Document Object Model**.
- It's the structure of your webpage represented as a **tree-like structure** (HTML elements are nodes).
- JavaScript can read and **change the DOM** to update the content, styles, etc.



▼ Why Use Virtual DOM?

- Directly changing the **real DOM is slow**, especially for large apps.
- When you update something, the browser checks and reloads the entire parts of the tree, which can slow down the performance.
- React solves this by using the Virtual DOM.

▼ What is Virtual DOM?

- A **lightweight copy** of the real DOM is called as Virtual DOM.

- And it is stored inside the memory.
- It allows React to manage changes more efficiently by minimizing the direct manipulation required by the Real DOM.
- Virtual DOM is a clone of the Real DOM.

▼ Diffing Algorithm

- React compares the **new Virtual DOM** with the **previous one**.
- It finds out **what has changed**.
- This comparison process is called **diffing**.

▼ Updating the Real DOM

- Once the diffing process identifies the necessary changes, React applies the changes to the Real DOM in a batched manner.
- After that, React knows what changed (using diffing), it updates **only those parts** of the real DOM.
- This is **much faster** than reloading the whole page or entire DOM.

▼ Patching

- Patching means **applying the changes** found during the diffing process.
- React sends only the **updated parts** to the real DOM.
- Once the diffing algorithm identifies the changes, React generates a set of patches - instructions for updating the Real DOM.
- These patches are applied in a batched manner, minimizing the number of DOM manipulations and improving performance.

▼ Reconciliation Process

- The process of comparing the previous Virtual DOM copy with the new one using a diffing algorithm is known as the **Reconciliation Process**.
- This is the **whole process** where:
    1. React builds a new Virtual DOM.

2. Compares it with the old one (diffing).
3. Find what changed.
4. Updates (patches) the real DOM.

▼ Steps of Reconciliation Process

1. React renders a new Virtual DOM.
2. It compares (diffing) it with the previous Virtual DOM.
3. Finds the changes.
4. Only the changed parts are updated (patched) in the real DOM.
5. The UI updates efficiently.

▼ Babel

- **Babel** is the tool that converts modern **JavaScript (ES6+)** and **JSX** into older JavaScript that browsers can understand.
- It is a **JavaScript Compiler** used to convert JavaScript (ES6+) code into a backwards-compatible version of JavaScript.
- It is internally converting JSX into browser-understandable JavaScript.
- For example, it changes this:

```
const element = <h1>Hello</h1>;
```

- Into this:

```
const element = React.createElement("h1", null, "Hello");
```